MIL-F-36911A
4 December 1975
SUPERSEDING
MIL-F-36911 (DSA-DM)
26 January 1973

MILITARY SPECIFICATION

FABRIC, NONWOVEN, BED SHEETS AND GOWNS, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type and eight colors of a nonwoven fabric for use in manufacture of disposable bedsheets and gowns.

1.2 Classification. The disposable, nonwoven fabric shall be furnished in the following colors, as specified (see 6.2):

White	Aqua	Yellow	Blue
Blue-Grey	Blue-Green	Misty-Green	Grey-Green

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

Federal

FED-TEST METHOD STD NO. 191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6532

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL FIRE CODES

702 Flammability of Wearing Apparel

(Application for copies should be addressed to the National Fire Protection Association, International (NFPA), 470 Atlantic Avenue, Boston, MA 02110.)

COMMERCIAL STANDARD (CS) 191

Flammability of Clothing Textiles

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20025.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Fabric. The nonwoven fabric shall be fabricated from any of the following compositions at the manufacturer's option. The finished nonwoven fabric shall be soft, have good draping and hand (feel). The finished nonwoven fabric shall not have an objectionable odor and shall be non-irritating and non-toxic when inspected as specified in 4.2.2.3. The finished fabric shall conform to the requirements of 3.1 and Table I when inspected as specified in 4.2.2 through 4.2.2.2.

a. The nonwoven fabric shall consist of 60 ± 5 percent bleached chemical pulp tissue, 15 ± 5 percent viscose rayon fibers (2 inch average staple length) and 25 ± 5 percent plastisol bonding and fire retardant agents. The fabric shall be constructed by laminating two layers of tissue having sandwiched between them two layers of oriented rayon fibers. One layer of the rayon fiber shall be placed in the cross direction.

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b. The nonwoven fabric shall consist of 40 ± 5 percent viscose rayon fibers and 60 ± 5 percent bleached chemical pulp tissue bonded with a resin adhesive and fire retardant agents. The fabric shall be made from a carded web or randomized web structure.

c. The nonwoven fabric shall consist of 45 ± 5 percent bleached chemical pulp tissue, 20 ± 5 percent synthetic fibers, 35 ± 5 percent polymeric bonding and fire retardant agents. The fabric shall be constructed by laminating two layers of nonwoven tissue having sandwiched between them one layer of random or oriented continuous synthetic filament fibers.

d. The nonwoven fabric shall consist of 100, +0, -5 percent viscose rayon fibers bonded with a resin adhesive and fire retardant agents. The fabric shall be an all-textile product made from a carded web or randomized web structure.

e. The nonwoven fabric shall consist of 87 ± 5 percent chemical pulp tissue, 1-1/2 ± 1/2 percent oriented continuous synthetic yarns in a nonwoven scrim and 11-=1/2 ± 5 percent bonding and fire retardant agents. The fabric shall be constructed by laminating two layers of tissue on each side of the scrim reinforcing network.

f. The nonwoven material shall consist of 60 ± 5 percent bleached chemical pulp tissue, 15 ± 5 percent rayon and polyester fiber blend (2 inc average staple length), and 25 + _ 5 percent plastisol bonding and fire retardant agents. The material shall be constructed by laminating two layers of tissue having sandwiched between them one layer of random or oriented fibers. One layer of the fibers shall be placed in the cross direction.

g. The nonwoven material shall consist of 60 ± 5 percent bleached chemical pulp tissue, 15 ± 5 percent synthetic fibers (2 inch average staple length), and 25 + _ 5 percent plastisol bonding and fire retardant agents. The material shall be constructed by one layer or by laminating two layers of nonwoven tissue having sandwiched between them two layers of oriented or random synthetic fibers. One layer of the synthetic fiber may be placed in the cross direction.

h. The fabric shall be nonwoven (saturation bonded wet laid) manufactured from cellulosic or synthetic fiber or a combination of both. The nonwoven fabric shall be resin saturated.

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Table I Nonwoven Fabric Properties

Characteristic	Requirements
Weight/Sq. Yd. Minimum	40 Grams or 1.12 ounces
Breaking Strength, Pounds Minimum:	
Machine Direction	2.6 1/ or 3.5 2/
Cross Direction	2.6 1/ or 1.0 2/
Flame Resistance, Seconds Minimum	8 3/ or 20 4/
Colorfastness to:	
Water 5/	Good
Crocking wet and dry 5/	Not lower than Munsel Value 8.0
Color	White, Aqua, Yellow, Blue, Blue-Grey, Blue-Green, Misty-Green or Grey-Green, as specified

1/ Applicable when breaking strength is determined by the grab method, Method 5100 of FED-TEST METHOD STD NO. 191.
2/ Applicable when breaking strength is determined by the cut strip method, Method 5102 of FED-TEST METHOD STD NO. 191.
3/ Applicable when flame resistance tests are conducted in accordance with NFPA 702.
4/ Applicable when flame resistance tests are conducted in accordance with CS 191.
5/ Applicable to all fabrics except white.

3.3 Workmanship. The finished sheet shall be clean, free from any break, tear, hole, wrinkle, uneven or ragged edge, foreign matter, spot, stain, mend, patch, and any other defect which detracts from its appearance or may impair its serviceability. The fabrics shall conform to the quality and grade established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of quality. When certificates of quality are submitted, they shall contain verifiable actual test and inspection data. The Government reserves the right to inspect and test the components to verify the validity of the certification.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (an investigation without the use of special laboratory apparatus or procedures such as visual) and testing (a determination by technical means of the physical and chemical properties) of the item.

4.2 Quality conformance inspection. The sampling and inspection shall conform to MIL-STD-105 except where otherwise indicated hereinafter.

4.2.1 Examination of the fabric. The examination of the fabric shall be in accordance with 4.2.1.1.

4.2.1.1 Yard-by-yard examination. The required yardage of each roll shall be visually examined and defects clearly noticeable at normal inspection distance (3 feet) shall be scored as classified in Table II. The defects found shall be scored regardless of their proximity to each other, except when two or more similar defects represent a single local condition of the fabric, in which case only the more serious defect shall be scored. A continuous defect shall be counted as one defect for each lengthwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be one linear yard. The sample size shall be in accordance with MIL-STD-105, Inspection Level II with an acceptable quality level of 4.0 defects per 100 units (yards) for major defects and 10.0 defects per 100 units (yards) for minor defects.

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Table II

Classification of Defects for the Visual Examination of the Finished Fabric

Defect	Major	Minor
GENERAL		
a. Any tear, hole, or cut through the material over 4 inches in the greatest dimension regardless of location	X	
b. Any tear, hole, or cut through the material over 1/2 inch but not greater than 4 inches in the greatest dimension regardless of location		X
c. Any tear, hole, or cut through the material 1/4 inch but not greater than 1/2 inch in the greatest dimension and is more than 10 inches from any edge		X
d. Any mend or patch	X	
e. Color not as specified	X	
f. Color not uniform or containing shade bars		X
g. Any edge that is ragged, on an angle or not straight		X
h. Single ply areas 1/:		
1. Straight gaps - a gap more than 1/2 inch wide at the widest point which runs across the entire width of the fabric	X	
2. Any single ply along any edge which is more than one inch wide at the widest point	X	
3. Triangular gaps on the interior - a gap more than 1/2 inch at the widest point	X	
4. Turned back corners - an area which is larger than 2 inches and larger than 4 inches on any adjacent sides of the triangle	X	

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Table II (contd)

Classification of Defects for the Visual Examination of the Finished Fabr

Defect	Major	Minor
CLEANNESS		
a. Any dark stain, grease, dirt, insect, clump of hair, metal, or other foreign matter greater than 2 inches in diameter	X	
b. More than 20 spots or streaks each greater than one square centimeter in area on the surface of the fabric		X
c. Any longitudinal pressed in or calendered wrinkle, pleat, or "Z" fold in the wadding greater than 1-1/2 inches in width and 12 inches in length		X
d. Objectionable odor beyond that expected in normal processing operations	X	
e. Any edge that is not clean for more than 6 inches		

1/ These are defects which do not go through the material and where the fiber is attached firmly to one ply.

4.2.2 Testing of the fabric. The fabric characteristics shall be determined in accordance with the inspection methods of FED-TEST METHOD-STD NO. 191 specified in Table III and 4.2.2.1 through 4.2.2.3. The test reports shall contain the individual values as well as the final result. The sample unit shall be two linear continuous yards. Each two yard sample shall be taken from a different roll. All samples shall be taken from the beginning of the roll. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specifie in 3.1 and Table I. The sample size (number of sample units) shall be in accordance with the following:

Lot Size (Yards)	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table III Nonwoven Fabric Properties

Characteristic	FED-TEST METHOD-STD NO. 191
Weight	5041
Breaking strength	5100 or 5102
Colorfastness to:	
Water	5630
Crocking	5651

4.2.2.1 Material. The material of the fabric shall be determined by any standard commercial method. A certificate of quality may be submitted (see 4.1.1).

4.2.2.2 Flame resistance. The flame resistance properties of the fabric shall be determined in accordance with NFPA 702 or CS 191.

4.2.2.3 Toxicity. The toxicity and irritability of the fabric shall be determined by any standard method. The test method used shall be approved by the contracting officer. A certificate of quality may be submitted for the non-toxic and non-irritating properties (see 4.1.1). Certification in lieu of testing is acceptable provided that the contractor certifies in writing that the raw materials and processing are identical to those used on fabrics previously tested and found acceptable.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. The fabrics shall be packaged and packed in a manner to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

6. NOTES

6.1 Intended use. The disposable, nonwoven fabrics covered by this specification are intended for use in the fabrication of disposable bed sheets and examination and isolation gowns.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.
b. Color and quantity desired (see 1.2).

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency -

Project No. 6532-1105
CS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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